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MTH:2310 DISCRETE MATH
QUIZ 9
DUE: MONDAY APRIL. 28

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(6 points) The Basics of Counting*
How many different 4-digit PIN's are there if no digit may be used more than once?

There are 10 digits 0–9

_ _ _ _

$10 \quad 9 \quad 8 \quad 7$

There are $10 \cdot 9 \cdot 8 \cdot 7$ total such PIN's.

Problem 2. *(6 points) The Basics of Counting*
How many bit strings are there of length at most 4?

_ _ _ _
$2 \; 2 \; 2 \; 2 \rightarrow 2^4$

_ _ _
$2 \; 2 \; 2 \rightarrow 2^3$

_ _
$2 \; 2 \rightarrow 2^2$

_
$2 \rightarrow 2^1$

There are 2 digits 0-1 to use in a bit string. A bit string of length at most 4 could be length 1, 2, 3, or 4 (no empty string)

There are $2^4+2^3+2^2+2$ many such bit strings

Problem 3. *(8 points) The Basics of Counting*
How many passwords are there which have exactly 12 characters and where each character may only be used once and must be one of the following

- a letter A - Z
- a letter a - z
- a digit 0 - 9
- a special symbol ! or @ or & or ?

There are 26 letters A–Z
" " " " a–z
" " 10 digits 0–9
" " 4 symbols ! @ & ?

$\Bigg\}$ There are a total of $26+26+10+4=66$ distinct characters

_ _ _ _ _ _ _ _ _ _ _ _
66 65 64 63 62 61 60 59 58 57 56 55

So there are $(66)(65)(64)(63)(62)(61)(60)(59)(58)(57)(56)(55)$ many such passwords.

Problem 4. *(8 points) Permutations and Combinations*
How many bit strings of length 8 are there with at least 4 zeros? Express your answer using the notation $P(n, r)$ or $C(n, r)$ or $\binom{n}{r}$. You do not need to simplify your answer.

At least 4 zeros could be 4, 5, 6, 7, or 8 zeros.

Choosing # of zeros is the same as choosing # of positions.

_ _ _ _ _ _ _ _
position: 1 2 3 4 5 6 7 8

4 zeros	5 zeros	6 zeros	7 zeros	8 zeros
$\binom{8}{4}$	$\binom{8}{5}$	$\binom{8}{6}$	$\binom{8}{7}$	$\binom{8}{8}$

So there are $\binom{8}{4}+\binom{8}{5}+\binom{8}{6}+\binom{8}{7}+\binom{8}{8}$ many such bit strings.

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Problem 5. *(12 points) Permutations and Combinations*

(a) *(6 points)* How many different two-digit sequences can be made with the numbers 1, 2, 3, 4? Express your answer using the notation $P(n,r)$ or $C(n,r)$ or $\binom{n}{r}$. List all of the sequences.

No repeated number.

Pulling from $n=4$ total distinct objects

Want groups of size $k=2$.

For sequences $\{1,2\} \neq \{2,1\}$ so order does matter

Then we want all 2-permutation from 4 digits.

There are $P(4,2) = \frac{4!}{2!}$ many such sequences.

$\{1,2\}. \{1,3\}. \{1,4\}. \{2,3\}. \{2,4\}. \{3,4\}$

$\{2,1\}. \{3,1\}. \{4,1\}. \{3,2\}. \{4,2\}. \{4,3\}$

(b) *(6 points)* How many different values can be made by ~~adding~~ multiplying exactly three numbers from 1, 2, 3, 4, 5? without repeating a digit Express your answer using the notation $P(n,r)$ or $C(n,r)$ or $\binom{n}{r}$. List all of the different values.

Pulling from a set of $n=5$ distinct objects.

Want groups of $k=3$.

When multiplying, $1\cdot2\cdot3 = 2\cdot1\cdot3$ so order does not matter.

So we want all 3-combinations of 5 objects.

There are $\binom{5}{3}$ different values.

Since no number 1-5 is the product of 2 distinct numbers 1-5 (except 1 and itself) then no two different groups of 3 numbers 1-5 can be equal. Each group gives a distinct product.

$1\cdot2\cdot3 = 6$	$1\cdot3\cdot4 = 12$	$2\cdot3\cdot4 = 24$	$3\cdot4\cdot5 = 60$
$1\cdot2\cdot4 = 8$	$1\cdot3\cdot5 = 15$	$2\cdot3\cdot5 = 30$	
$1\cdot2\cdot5 = 10$	$1\cdot4\cdot5 = 20$	$2\cdot4\cdot5 = 40$	

Problem 6. *(14 points) Permutations and Combinations*

(a) *(8 points)* How many different ways can exactly 5 strictly positive integers be added together to get 20 **if the order of the sum matters**? That is, $1+1+1+1+16$ is taken to be different from $1+1+1+16+1$. Express your answer using the notation $P(n,r)$ or $C(n,r)$ or $\binom{n}{r}$. You do not need to simplify your answer.

$$20 = \overbrace{(1+1+1+1+1)}^{5} \overset{+}{+} \overbrace{(1+1+1+1+1+1+1)}^{7} \overset{+}{+} \overbrace{(1+1+1+1+1+1+1+1)}^{8}$$

Every such sum can be found by choosing 3 distinct groups (setting parentheses).

To create 3 distinct groups, we mark 2 out of 19 "+" signs as the ones between groups.

There are $\binom{19}{2}$ many such sums.

(b) *(6 points)* Let N and k be positive integers with $k \leq N$. How many different ways can exactly k strictly positive integers be added together to get N **if the order of the sum matters**? Express your answer using the notation $P(n,r)$ or $C(n,r)$ or $\binom{n}{r}$.

Using the same strategy, if I want K groups, I mark $(K-1)$ "+ signs" to separate them. If I have N 1's, that leaves $N-1$ total "+ signs"

There are $\binom{N-1}{K-1}$ total such sums

for any integers $N \geq K \geq 1$.

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Problem 7. *(14 points) Pigeonhole Principle*

(a) *(6 points)* Suppose exactly six integers are chosen at random. Is it necessarily true that at least two of them will have the same remainder when divided by 5? Justify your answer.

There are exactly 5 different remainders when dividing by 5.

Then there is 1 more integer than possible results, so by standard pigeonhole principle it is true that at least two must have the same remainder when divided by 5.

(b) *(8 points)* Suppose exactly N integers are chosen at random. If possible, what is the **smallest** value of N that guarantees that at least four of them will have the same remainder when divided by 5? Justify your answer. If there is no such integer N, explain why.

$N = ?$

$K = 5$

$$\left\lceil \frac{N}{k} \right\rceil = 4$$

Then by generalized pigeonhole principle if we want the smallest N such that $\left\lceil \frac{N}{5} \right\rceil = 4$, N must be 21